

Attack–Fault Ambiguity Against a Strategic Attacker: The Stealth–Harm Frontier, Exploitable Commitments, and the Verification Margin

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Abstract—

Draft pending. Claim: against a committed belief-based operator, an ε -stealthy attacker faces a stealth–harm frontier; exact stealth is generically impossible under state feedback, and verification shifts the frontier against the attacker.

I. INTRODUCTION

Planned. Papers 1–3 treat the attacker as fixed; a leader–follower formulation answers whether the framework survives an adversary who knows the rule.

II. RELATED WORK

Planned. Stealthy attacks in cyber-physical systems, security games, adversarial classification.

III. MODEL

Planned. Stackelberg game; attacker constrained to total-variation distance ε from the fault law. Code: `src/paper4_game`.

IV. THE STEALTH–HARM FRONTIER

Planned. Stealth–harm frontier $\mathcal{F}_W(\varepsilon)$ (Definition G9); harm ceiling as the $\varepsilon \rightarrow 0$ limit (Theorem 1 output-level; zero excess harm generically under Theorem 2).

V. NUMERICAL ILLUSTRATION

Planned.

VI. DISCUSSION

Planned.

VII. CONCLUSION

Planned.

REFERENCES

- [1] F. Pasqualetti, F. Dörfler, and F. Bullo, “Attack detection and identification in cyber-physical systems,” *IEEE Transactions on Automatic Control*, vol. 58, no. 11, pp. 2715–2729, 2013.
- [2] T. Roy and S. Dey, “On distinguishability of anomalies as physical faults or actuation cyberattacks,” *ASME Letters in Dynamic Systems and Control*, vol. 4, no. 3, p. 031003, 2024, arXiv:2311.09342.